

## Michael Dawson-Haggerty

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| CONTACT    | (631) 662-1266                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | mikedh@kerfed.com |
| INTERESTS  | I'm interested in making mechanical problems “just a software problem” through physical automation and software. My primary technical interests are machine design and computational geometry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                   |
| EDUCATION  | <b>Carnegie Mellon University</b> , Pittsburgh, PA<br>MS Mechanical Engineering, 2011 <ul style="list-style-type: none"><li>• Worked on a flexible fixturing grant from ABB Robotics under Prof. David Bourne.</li></ul> <b>Tufts University</b> , Medford, MA<br>BSME Mechanical Engineering, 2010 <ul style="list-style-type: none"><li>• <i>Senior Project</i>: Designed and fabricated capacitive sensor to measure a 0.001-0.012” gap using vacuum sputtered gold on a Mylar membrane under Dr. Robert White and Dr. Chris Rogers.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |
| EXPERIENCE | <b>Kerfed Intelligent Manufacturing</b> , Pittsburgh, PA<br><i>Robotics Engineer, Founder</i> November 2017 - Present <ul style="list-style-type: none"><li>• Wrote and maintain <a href="#">trimesh</a>, a popular 30M+ download open source Python geometry library deemed a “critical project on PyPi” and is a dependency for Google’s Tensorflow-Graphics, Nvidia’s Kaolin, Cura (the most popular slicer for 3D printing), and many others. Trimesh is used by Berkeley’s Autolab, CMU, many universities, NASA, Autodesk, Apple, etc: <a href="https://github.com/mikedh/trimesh">https://github.com/mikedh/trimesh</a></li><li>• Wrote most of Kerfed’s proprietary manufacturing stack which takes assemblies from CAD systems, de-duplicates, automatically detects from geometry if each part is flat, bent, rolled, pocket milled, turned, or COTS (geometric search) without requiring any manual analysis: <a href="https://youtu.be/slfWvma1s_M">https://youtu.be/slfWvma1s_M</a></li><li>• Founded company and launched successful manufacturing service <a href="https://kerfed.com">https://kerfed.com</a> and API provider <a href="https://kerfed.dev">https://kerfed.dev</a></li><li>• Managed project and developed components for a complex second-generation analysis system based on temporal.io and protobuf3 with WGSL shader volumetric milling analysis: <a href="https://carviewizard.com">https://carviewizard.com</a>.</li></ul> <b>Berkshire Grey</b> , Pittsburgh, PA<br><i>Robotics Engineer (Employee #7)</i> January 2015 - November 2017 <ul style="list-style-type: none"><li>• On the founding team of Berkshire Grey, a company founded by the former CTO of iRobot and backed by Khosla, NEA, and Softbank that uses mobile robots and robotic manipulation for warehouse automation. Company went public as BGRY on the NASDAQ.</li><li>• Met with teams from Walmart, FedEx, Target and other undisclosed customers at dozens of different distribution centers and worked with BGRY’s team to build and deploy robotic systems to meet customer needs.</li></ul> |                   |

- Designed many complete electromechanical systems from concept to production, including approximately a thousand unique parts: welded steel structures, flat (waterjet or laser) parts, bent (CNC press brake) parts, and machined parts (milled or turned). Designed most mechanical systems and parts in this video: [https://youtu.be/\\_yrQTdVFcuU](https://youtu.be/_yrQTdVFcuU)
- Handled supplier identification and relationships for millions of USD worth of mechanical components. Planned and executed “low rate initial production” line for mobile robots at Segue, a contract manufacturer in Lowell MA.
- 15+ patents: <http://patents.justia.com/inventor/michael-dawson-haggerty>

## **Carnegie Mellon University Robotics Institute, Pittsburgh, PA**

*Systems Software and Hardware Engineer*

May 2011 - January 2015

- Worked in the Manipulation Lab under Dr. Matt Mason (former director of CMU’s Robotics Institute), the Personal Robotics Lab under Dr. Sidd Srinivasa (Director of Robotics at Amazon) and the Rapid Manufacturing Lab under David Bourne.
- Designed and integrated computing and sensors into compact custom unit for Foxconn factory health project.
- Executed successful upgrade program on HERB the bimanual robot (funded by NSF and ONR), including new frame with integrated 4U rack, custom battery, 1U compute unit (including a full size GPU in 1U), and redesigned power distribution system.
- Designed and integrated cell for Acutronic USA under a Research for Advanced Manufacturing grant to automate IMU calibration by parallelizing thermal soak.
- Authored software to integrate the iRobot-Harvard hand (later developed into RightHand Robotics) for CMU-NREC’s ROS based software environment for DARPA ARM-S.
- Designed, built, and integrated pan-tilt unit for HERB, featuring integrated joint encoders, custom slip clutch, custom electromechanicals.
- Principal engineer on CMU’s branch of DARPA AVM-iFAB: rapid geometry-based fabrication of tubular welded structures. Designed and sourced ABB IRB2600 work cell, HMMWV frame fixture, robot mounted sensor package, sensor interface electronics, and robot-DAQ synchronization. Wrote all non-vision cell software.

## **Tufts University, Medford, MA**

*ITS Linux Support*

August 2007 to May 2010

- Support faculty Linux users, including setting up Linux workstations and getting builds of various research packages to compile.

## **Brookhaven National Labs, Upton, NY**

*SULI Undergraduate Research Intern*

Summer 2007, Summer 2008

- Developed tooling and software to bring the tiled CCD focal plane of the LSST telescope (now called [Vera Rubin](#)) to a target peak-valley flatness of  $5\mu\text{m}$  across a 12cm 3x3 grid.

MISCELLANEOUS **EIT NH #5373 (successfully completed Mechanical FE)**

June 2010

Video demos: <https://youtube.com/mik3dh>

Code: <https://github.com/mikedh>